

A Grammatical Sketch of Turkish

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Contents

1	Introduction	4
2	Sound System	6
2.1	Inventory of Phones	6
2.1.1	Consonant Chart	6
2.1.2	Vowel Chart	7
2.2	Contrastive Features	7
2.2.1	Voicing	7
2.2.2	Vowel Length	8
2.3	Phonemic Inventory & Allophony	8
2.4	Phonotactic Constraints	10
2.5	Vowel Harmony & Consonant Assimilation	12

3 Morphology	14
3.1 Inflectional Suffixes	14
3.2 Derivational Suffixes	16
3.3 Order of Suffixes	17
3.4 Allomorphy	17
4 Syntax	18
4.1 SOV Word Order	18
4.2 Head Directionality Analysis	19
4.2.1 Adpositional Phrases	19
4.2.2 Noun Phrases	20
4.2.3 Verb Phrases	21
4.2.4 Complementizer Phrases	21
4.2.5 Overall Head Directionality	22
4.3 Interrogative Sentences	23
4.3.1 Yes/No Questions	23
4.3.2 Wh-Questions	24
5 Research Proposal	25
6 Summary & Conclusion	27
7 Appendix	29
7.1 Free Morphemes	29
7.2 Bound Morphemes	33
References	35

Disclaimer

This paper was written as part of an introductory linguistics course at MIT (i.e. 24.900: Introduction to Linguistics) which I took in Spring 2026. In accordance with the course guidelines, all analyses presented here are based solely on my independent fieldwork conducted with my language consultant, without consulting any external sources. While I have made every effort to ensure the analyses are as accurate as possible, this paper inevitably reflects my own observations and interpretations of the data collected and may therefore contain errors and inconsistencies. As such, this paper should be regarded as an educational work rather than an authoritative academic reference.

Acknowledgment

I would like to express my sincere gratitude to Şebnem, my language consultant, for her patience, generosity, and willingness to share her language throughout this project. Her enthusiasm, thoughtful explanations, and collaboration made this fieldwork both possible and deeply enjoyable. I am truly grateful for the time and effort she devoted to helping me in this project.

1 Introduction

This paper provides a brief overview of the phonetic, phonological, and morphological properties of the Turkish language based on data I collected through fieldwork.

According to Omniglot [1], Turkish is a member of the Oghuz branch of the Turkic language family, spoken by approximately 88 million people worldwide. It is mainly spoken in Turkey, but also in other regions such as Northern Cyprus, Cyprus, Greece and Iraq. Historically, Turkish was once written in a version of the Arabic script, but was later replaced with a version of the Latin alphabet in 1928, as part of a language reform aimed to modernize Turkey.

My fieldwork consultant, Şebnem, lived in Ankara, Turkey for the first 15 years of her life and speaks Turkish natively. In addition to acquiring Turkish naturally from her environment, she was also exposed to the language in a formal setting, as Turkish served as the primary language of instruction in her school. She reports relatively low dialectal variation within Turkey, though she notes that speakers from certain regions, like those from Kurdish communities, may exhibit slight accent differences.

Over the past three months, I met with Şebnem on four main occasions for dedicated fieldwork sessions in the form of verbal discussions. Because she is also someone whom I interact with regularly, it was easy for me to ask follow-up questions and seek clarification outside of scheduled sessions.

Before sessions, I would typically first prepare a list of English words or phrases, depending on the specific linguistic aspect I aimed to investigate.

Then, during the sessions, Şebnem would elicit the corresponding Turkish translations, and I would record them down. In some cases, I would instead ask more structured phonological questions, such as requesting sets of words containing a specific sound or minimal pairs differing by a single sound. Accordingly, Şebnem would then provide the relevant Turkish words along with their English translations.

During the sessions, I would take handwritten notes to record Şebnem's responses. When working on morphology or syntax, I would simply write down my best approximation of the Turkish word in conventional orthography. On the other hand, if I was focused on the phonetic or phonological details, I would refer to an online IPA chart to write out an IPA transcription of her response. I would also frequently ask follow-up questions and repeat words back to her to ensure an accurate transcription. Additionally, I audio-recorded our sessions with her consent, which allowed for later review when necessary.

In this paper, I summarize my findings on Turkish phonetics, phonology, and morphology as observed through my fieldwork. Based on my data collection approach, transcriptions in Section 2 (Sound System) use IPA transcriptions, while Sections 3 (Morphology) and 4 (Syntax) use my best approximation of Turkish spelling in conventional orthography. Phrases are glossed according to Leipzig Glossing Rules [2].

2 Sound System

This section discusses the inventory of phones and phonemes found in Turkish, followed by phonological constraints that can be proposed based on the data. Additionally, it briefly highlights two intriguing phonological processes of Turkish, known as vowel harmony and consonant assimilation.

2.1 Inventory of Phones

Presented below are all the sounds I've encountered throughout my fieldwork.

2.1.1 Consonant Chart

	Bilabial	Labio-dental	Alveo-lar	Post-alveolar	Palatal	Velar	Glottal
Plosive	p ^h b		t ^h d			k ^h g	
Nasal	m			n			
Tap/ Flap			ɾ r				
Fricative		f v	s z	ʃ		ɣ	h
Liquid			l				
Glide					j		
Affricate				tʃ dʒ			

Table 1: Consonant Sounds in Turkish

Note: Where symbols appear in pairs, the one to the left represents an unvoiced consonant, while the one to the right represents a voiced consonant.

Turkish has a moderately rich consonant inventory of around 20 sounds. Consonant sounds in Turkish have close equivalents in English, with the exception of the velar fricative [ɣ]. Furthermore, I noticed that voiceless plosives

are always aspirated in Turkish (and is especially noticeable at the end of words), implying that aspiration is not contrastive.

2.1.2 Vowel Chart

	Front	Central	Back
High	i/ɪ	ʉ	ʊ/ u / ʊ
Mid	ɛ/e/ œ /ø	ə	o /ɔ
Low	a/a:	ɐ	-

Table 2: Vowel Sounds in Turkish

Note: Vowels in bold are rounded.

While I identified many vowel sounds here, only a subset of them are phonemes, as discussed in Section 2.3.

2.2 Contrastive Features

2.2.1 Voicing

Voicing in Turkish is contrastive.

Here are some minimal pairs that show this:

[k <u>u</u> s] ‘down/decrease’	[k <u>u</u> z] ‘girl’
[t <u>u</u> mɛn] ‘group of’	[d u <u>m</u> ɛn] ‘ship steering wheel’
[b ak] ‘look’	[p <u>a</u> k] ‘clean’
[sa <u>f</u>] ‘pure’	[sa <u>v</u>] ‘thesis’
[k az] ‘goose’	[g <u>a</u> z] ‘gas’

Table 3: Minimal Pairs Showing Voicing is Contrastive

Accordingly, /p/ and /b/ are analyzed as separate phonemes, along with the other voiceless-voiced pairs.

2.2.2 Vowel Length

Additionally, vowel length is also contrastive in Turkish, though such contrasts are rare. Here are some minimal pairs:

[kan] ‘blood’	[ka:n] ‘Kaan (a name)’
[hala] ‘aunt’	[ha:la] ‘still’
[kaɾ] ‘snow’	[ka:ɾ] ‘profit’
[ama] ‘but’	[a:ma:] ‘blind’
[adɛt] ‘singular’	[a:ɛt] ‘custom/tradition’

Table 4: Minimal Pairs Showing Vowel Length is Contrastive

Interestingly, the only long vowel that seems to appear in Turkish is a long [a] (denoted by â in Turkish script). According to my consultant, most (if not all) of the words containing a long [a] are loan words. Also, my consultant reports that the long [a] slightly affects the pronunciation of adjacent consonants, though she is unable to specify exactly how.

2.3 Phonemic Inventory & Allophony

Based on the contrasts I’ve identified, I propose a phonemic inventory consisting of 20 consonant phonemes and 8 vowel phonemes. The following tables display the proposed phonemic inventory, as well as an example of a word in which the phoneme appears.

/p/ ([køpɛk], ‘dog’)	/b/ ([badʒak], ‘leg’)	/t/ ([at], ‘horse’)
/d/ ([kɛde], ‘cat’)	/k/ ([køpɛk], ‘dog’)	/g/ ([gaz], ‘gas’)
/m/ ([masa], ‘table’)	/n/ ([anːɛ], ‘mother’)	/r/ ([farɛ], ‘mouse’)
/f/ ([farɛ], ‘mouse’)	/v/ ([tavuk], ‘chicken’)	/s/ ([masa], ‘table’)
/z/ ([aʏaz], ‘mouth’)	/ʃ/ ([ɛʃɛk], ‘donkey’)	/ʎ/ ([aʏaz], ‘mouth’)
/h/ ([hala], ‘aunt’)	/l/ ([ɛl], ‘hand’)	/j/ ([ije], ‘good’)
/tʃ/ ([atʃ], ‘hungry’)	/dʒ/ ([badʒak], ‘leg’)	

Table 5: Phonemic Inventory of Consonants in Turkish

	Front	Central	Back
High	/i/ ([ije], ‘good’)	/ʉ/ ([kʉtʃʉk], ‘small’)	/ʉ/ ([kʉz], ‘girl’) /u/ ([uju], ‘sleep’)
Mid	/ɛ/ ([anːɛ], ‘mother’) /ø/ ([køpɛk], ‘dog’)		/o/ ([kol], ‘arm’)
Low	/a/ ([hala], ‘aunt’)		

Table 6: Phonemic Inventory of Vowels in Turkish

In general, consonants in Turkish are pronounced consistently across different environments. The only variation I noticed concerns the phoneme /r/, whose pronunciation is contextually predictable. Consider the following words which contain the phoneme /r/:

- (1) a. [farɛ] ‘mouse’, [kardɛʃ] ‘sibling’, [øːrɛnɛk] ‘learn’
b. [biɾ] ‘one’, [kaɾ] ‘snow’, [bilɛsajar] ‘computer’

In (1a) where the phoneme /r/ is word-medial, it is typically pronounced as an alveolar tap, as represented by [r]. In contrast, when the phoneme /r/ is word-final as seen in (1b), the sound produced is softened and accompanied by an audible release of air, which I believe is best transcribed as [ɾ]. Despite

this phonetic variation, my consultant reports no difference in meaning when the pronunciation of /r/ is altered. This suggests that these two sounds are allophones of a single phoneme /r/.

Analyzing vowels now, the chart reveals that Turkish shows a preference for front or high vowels. Notably, [o] is the only vowel phoneme in the inventory that is neither front nor high. Vowels also exhibit greater variability in sound, unlike consonants. While I identified only eight vowel phonemes in Turkish, I observed many more vowel phones as shown in Table 2, which are realized across different contexts. By pronouncing alternative vowel sounds in words to my consultant, I was able to confirm with her that many of the phones I initially identified were non-contrastive and simply allophones. For example, [i], [ɪ], and [e] function as allophones of the single phoneme /i/. Despite this variability, it is generally not difficult to determine which phoneme is intended, though I did not deeply investigate the phonological rules for predicting vowel allophones. My consultant also noted that such variation may be influenced by accent or dialect.

2.4 Phonotactic Constraints

Based on my data, I propose several phonotactic constraints.

First, I noticed that vowels are generally not seen consecutively in Turkish, leading me to hypothesize that the language does not allow diphthongs. In some cases, vowels may come together, but they are pronounced as two separate vowels rather than one smooth sound. For example, in [an:εan:ε] (meaning grandmother), the [ε] and [a] come together but they are pro-

nounced one after the other, not as a diphthong.

Additionally, my hypothesis is further supported by cases of epenthesis of a glide in the merge of two morphemes, in which the first ends with a vowel and the second starts with a vowel. Consider the following:

- (2) a. [ø:rɛn-ɛdʒɛk]
 learn-FUT
 ‘He/She will learn’
- b. [ji-jɛdʒɛk]
 eat-FUT
 ‘He/She will eat’

In (2b), a glide [j] is inserted before the suffix [-ɛdʒɛk]. This suggests the glide ([j]) acts as a transition between the two vowels (that is, the [i] from [ji] and the [ɛ] from [-ɛdʒɛk]), to prevent them from occurring consecutively.

To confirm or refute my hypothesis, I should look at more cases where adding a suffix to a word (without modifying the suffix) would theoretically force two vowels together, and observe whether or not a glide would be inserted between the two vowels.

Second, I propose that word-final voiced stops are not allowed in Turkish. This is supported by the fact that I’ve not encountered any words in my data that end in a voiced plosive. Notably, there are words that end in voiceless stops (e.g. [kitap] ‘book’) and voiced consonants (e.g. [aɣaz] ‘mouth’), suggesting that the restriction is specific to voiced stops rather than stops or voiced consonants in general. However, further data would be required to determine whether this is a strict constraint or simply a strong tendency.

2.5 Vowel Harmony & Consonant Assimilation

Two notable phonological processes of Turkish are vowel harmony and consonant assimilation. These processes play an important role in determining the form of morphemes, especially in suffixes that attach to a root.

Broadly speaking, vowel harmony can be observed through the alternation of vowels in morphemes. For example, in my data, possible vowels for a suffix appear in sets, either $\{[e], [a]\}$ or $\{[i], [\text{ɯ}], [u], [\text{ʉ}]\}$.

Below are examples of vowel harmony at play. The first set illustrates the alternation of the suffix $[-m\epsilon]/[-ma]$, and the second set illustrates the alternation of $[-d\text{ʒ}i]/[-d\text{ʒ}\text{ɯ}]/[-d\text{ʒ}u]/[-d\text{ʒ}\text{ʉ}]$.

- (3) a. $[jap-ma]$ ‘don’t do’, $[git-m\epsilon]$ ‘don’t go’, $[konu\text{ʃ}-ma]$ ‘don’t speak’
 b. $[ʃarku\text{ɹ}-d\text{ʒ}\text{ɯ}]$ ‘singer’, $[kitap-tʃ\text{ɯ}]$ ‘bookseller’, $[futbol-d\text{ʒ}u]$ ‘footballer’

Initially, I hypothesized that the vowel of a suffix is determined by selecting the vowel that is most similar to the vowel in the preceding syllable, often matching in backness and roundedness. While this accounts for many cases, a notable inconsistency arises from my proposed vowel chart, where I identified $[a]$ as a front vowel. I noticed that if $[a]$ were instead analyzed as a back vowel, this hypothesis would account for a greater number of cases. However, even so, this generalization does not consistently account for all forms in the data. Take the following case for example:

- (4) a. $*[j\epsilon-m\epsilon-jor-\text{ʊ}m]$
 b. $[j\epsilon-mi-jor-\text{ʊ}m]$
 eat-NEG-PROG-1SG
 ‘I am not eating’

I predicted that the English sentence ‘I am not eating’ would translate to (4a) with the suffix [-mɛ], based on patterns I have identified. However, my consultant provided the correct translation as shown in (4b) with the suffix [-mi], which was inconsistent with the rules I thought I had established. When asked to explain this, my consultant was also unable to provide a clear explanation, suggesting the system is either not fully predictable or more complex than my initial hypothesis captures.

Furthermore, vowel harmony alternations do not just affect suffix vowels, but also vowels of roots.

- (5) a. *[jɛ-jor]
 b. [ji-jor]
 eat-PROG
 ‘He/She is eating’

For instance, the word ‘eat’ is [jɛ] in its bare form. However, if we add the suffix [-jor], the root appears as [ji] as in (5b), instead of [jɛ] as in (5a).

I also observed consonant assimilation in some instances, though less frequently. For example, the suffix [-dʒi] can also take the form [-tʃi], and the suffix [-diŋ] can also take the form [-tiŋ]. From the data available, consonant assimilation appears to involve voicing alternations, such as alternations between [t]–[d], [p]–[b], and [tʃ]–[dʒ]. This suggests that the voicing of a consonant may be influenced by its phonological environment, though further data would be needed to establish a consistent pattern.

3 Morphology

This section presents a brief introduction to the morphology of Turkish. In particular, an interesting morphological feature of Turkish is its heavy use of suffixation, which will be the focus of this section. In fact, based on the data I’ve gathered, I observed only suffixes and did not identify any prefixes or infixes. While this observation is limited by the scope of the dataset, I hypothesize that Turkish morphology relies exclusively on suffixation.

3.1 Inflectional Suffixes

In Turkish, there are inflectional suffixes that attach to a verb to indicate tense and subject agreement. As such, a verb in Turkish, once combined with these suffixes, can stand alone as a complete sentence. As a matter of fact, based on the sentences I’ve observed, it seems that verbs never occur in isolation, but instead require taking a suffix, unless in its imperative form.

	PAST	PROG	FUT
1SG	<i>yedim</i> ye-di-m eat-PAST-1SG ‘I ate’	<i>iyorum</i> ye-yor-um eat-PROG-1SG ‘I am eating’	<i>iyeyeceğim</i> ye-ecek-im eat-FUT-1SG ‘I will eat’
2SG	<i>yedin</i> ye-di-n eat-PAST-2SG ‘You ate’	<i>iyorsun</i> ye-yor-sun eat-PROG-2SG ‘You are eating’	<i>iyeyeceksin</i> ye-ecek-sin eat-FUT-2SG ‘You will eat’
3SG	<i>yedi</i> ye-di eat-PAST ‘He/she ate’	<i>iyor</i> ye-yor eat-PROG ‘He/she is eating’	<i>iyecek</i> ye-ecek eat-FUT ‘He/she will eat’

Table 7: Morphological Paradigm: Tense $\times$ Subject

Based on the data above, in the case of the verb *ye-* ('eat'), I identify the following tense markers: *-di* for past tense, *-yor* for progressive aspect, and *-ecek* for future tense.

Subject agreement is then expressed through additional suffixation.

For first person singular, the verb typically ends in *-m*, with a preceding vowel (determined by vowel harmony) if the tense suffix ended in a consonant.

For second person singular, the verb takes the suffix *-n*, preceded by the letter 's' and a vowel (determined by vowel harmony) if the tense suffix ended in a consonant (e.g. *-sin*).

For third person singular, there is no subject agreement suffix, and the subject defaults to 'he' or 'she'.

Note that the subject can also be made explicit by mentioning it before the verb. While a more detailed discussion of this falls under syntax, consider the following example:

(6) *Ali yiyor*

Ali *ye-yor*

Ali eat-PROG

'Ali is eating'

Thus, the general morphological order observed for verbs is

verb stem + tense/aspect suffix + subject agreement suffix (if present)

3.2 Derivational Suffixes

In addition to inflectional suffixes, Turkish also has derivational suffixes that attach to a root to form a new word or modify its part of speech.

For instance, consider the suffix -me/-ma which attaches to a verb to form its negative imperative form:

- (7) a. ye-me
eat-NEG
'Don't eat'
- b. git-me
go-NEG
'Don't go'
- c. uyu-ma
sleep-NEG
'Don't sleep'
- d. konuş-ma
speak-NEG
'Don't speak'

Another derivational suffix is -ci/-cı/-cü/-cu which attaches to a noun to refer to a person who does something related to that base noun:

- (8) şarkı-cı
song-A
'singer'

3.3 Order of Suffixes

Consider the following sentence:

- (9) ye-mi-yor-um
eat-NEG-PROG-1SG
'I am not eating'

From this, we conclude that the order of suffixes attaching to verbs is

verb stem + negation + tense/aspect + subject agreement

as similarly discussed in Section 3.1.

As for derivational suffixes, I did not investigate the ordering of multiple derivational suffixes, and thus do not have any patterns to report.

3.4 Allomorphy

Due to vowel harmony and consonant assimilation as discussed in Section 2.5, allomorphy is common in Turkish. That is, a single morpheme has multiple forms, which, in the case of Turkish, largely depends on its phonological environment. For example, as shown in Section 3.2, the negative suffix appears as either -me or -ma. Importantly, these two morphemes carry the exact same meaning. This allomorphy is not unique to -me and -ma, but apply more broadly to other suffixes in Turkish as well.

4 Syntax

This section provides an overview of Turkish syntax. In particular, basic word order, head directionality, and the formation of interrogative sentences are analyzed.

4.1 SOV Word Order

In transitive sentences, Turkish follows SOV (Subject-Object-Verb) order.

- (10) a. Ben pilav yi-yor-um.
1SG rice eat-PROG-1SG
'I am eating rice.'
- b. Ben sen-i sev-iyor-um.
1SG 2SG-ACC love-PROG-1SG
'I am loving you.'
- c. Kedi kitap oku-yor.
cat book read-PROG
'The cat is reading a book.'

The data presented above support this claim. However, note that in sentences where the subject is a pronoun, such as in (10a) and (10b), it is acceptable (and more common) to omit the first subject pronoun (e.g. 'Ben', 'Sen', 'O', etc.) since it is already indicated by a suffix attached to the verb. For instance, the sentence below is valid and exactly equivalent to the sentence in (10a).

- (11) Pilav yi-yor-um.
rice eat-PROG-1SG
'I am eating rice.'

4.2 Head Directionality Analysis

Determining head directionality in Turkish is not entirely straightforward due to its complex morphology. Note that for the purposes of this analysis, suffixes are not considered to affect head directionality.

4.2.1 Adpositional Phrases

Turkish is head-final for adpositional phrases.

- (12) a. Sen-in ile yi-yor-um.
 2SG-GEN with eat-PROG-1SG
 ‘I am eating with you.’
- b. Ev-im-de yi-yor-um.
 house-1SG.POSS-LOC eat-PROG-1SG
 ‘I am eating in my house.’
- c. Ev-im-de sen-in ile yi-yor-um.
 house-1SG.POSS-LOC 2SG-GEN with eat-PROG-1SG
 ‘I am eating with you in my house.’
- d. Yatak-ta uyu-yor-um.
 bed-LOC sleep-PROG-1SG
 ‘I am sleeping on a bed.’

As shown in the data above, Turkish uses postpositions, in which words like ‘with’ or ‘in’ follow after its complement. This contrasts with English, which uses prepositions. In particular, a common postposition found in Turkish is the locative case suffix *-da/-de/-ta/-te*. As shown in (12b)-(12d), this suffix attaches to the end of the noun phrase it concerns. Hence, the use of postpositions in Turkish support the claim that adpositional phrases are head-final.

4.2.2 Noun Phrases

Noun phrases in Turkish are also head-final.

- (13)
- a. Kedi-ler hakkında bir kitap
cat-PL about one book
'A book about cats'
 - b. Oku-duğ-um büyük kitap
read-REL-1SG big book
'A big book that I read'
 - c. Oku-duğ-um kitab-ı sev-iyor-um.
read-REL-1SG book-ACC love-PROG-1SG
'I am loving the book that I read.'
 - d. Büyük kitap
big book
'A big book'
 - e. Bir büyük kitap
one big book
'One/A big book'
 - f. Büyük kitab-ım
big book-1SG.POSS
'My big book'
 - g. Ali'nin kitab-ı
Ali-GEN book-3SG.POSS
'Ali's book'

The examples above illustrate that complements (e.g. complementizer phrases) and even modifiers (e.g. adjectives) of a noun precede the noun, suggesting that Turkish noun phrases are head-final. In contrast, grammatical markings representing possession or determiners (e.g. 'my', 'the') are added as suffixes to the noun, as seen in (13c), (13f) and (13g).

4.2.3 Verb Phrases

Verb phrases in Turkish are also head-final. As presented in example (10) in Section 4.1, the verb consistently follows its complement (i.e. the object it takes), consistent with SOV word order. The verb is also typically marked by suffixes indicating tense and subject agreement.

4.2.4 Complementizer Phrases

Complementizer phrases are also head-final.

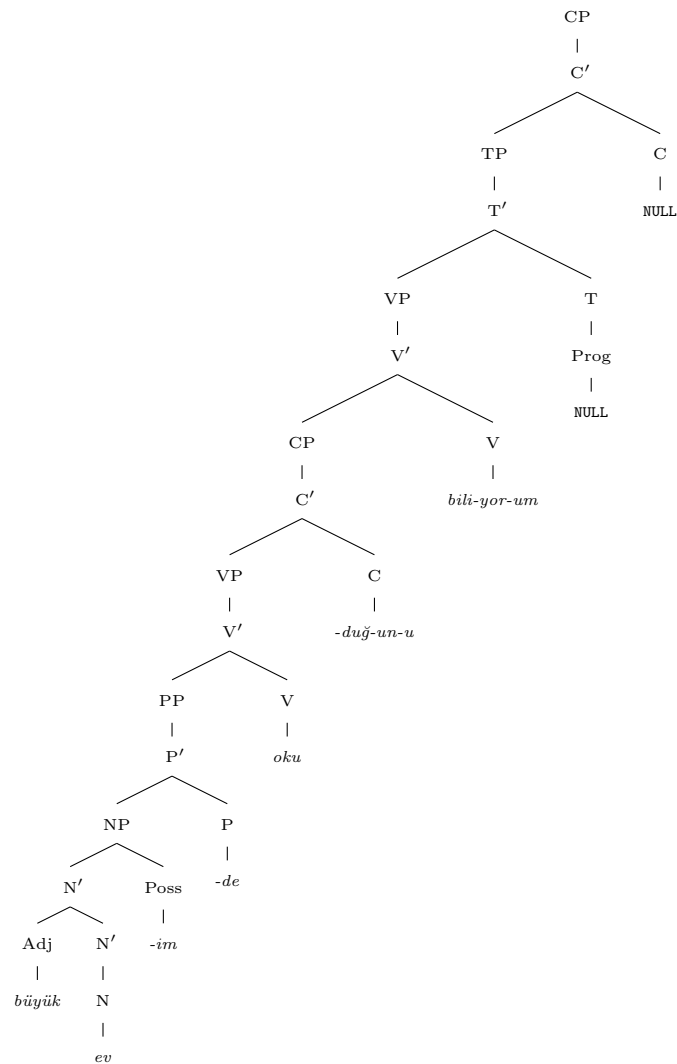
- (14) a. Havalı ol-duğ-un-u düşün-üyor-um.
cool be-REL-2SG-ACC think-PROG-1SG
'I am thinking that you are cool.'
- b. Oku-duğ-un-u bili-yor-um.
read-REL-2SG-ACC know-PROG-1SG
'I am knowing that you are reading.'
- c. Kedi-nin küçük ol-duğ-u-nu düşün-üyor-um.
cat-GEN small be-REL-3SG-ACC think-PROG-1SG
'I am thinking that the cat is small.'

These examples suggest that *duğ* is a complementizer. Indeed, the morpheme follows its complement (i.e. a verb phrase), followed by subject agreement and accusative suffixes.

4.2.5 Overall Head Directionality

Overall, the data suggests that Turkish is a head-final language, unlike English which is head-initial. The tree below illustrates the head-final structure of the following sentence.

- (15) Büyük ev-im-de oku-duğ-un-u bili-yor-um.
 big house-1SG.POSS-LOC read-REL-2SG-ACC know-PROG-1SG
 ‘I know that you are reading in my big house.’



4.3 Interrogative Sentences

4.3.1 Yes/No Questions

Turkish uses yes/no question markers, which are *mi/mı/mu/mü*, as determined by vowel harmony. Examples of its use are shown in the table below.

#	Statement	Yes/No Question
1	Yi-yor-um. eat-PROG-1SG 'I am eating.'	Yi-yor mı-yum? eat-PROG Q-1SG 'Am I eating?'
2	Yi-yor-sun. eat-PROG-2SG 'You are eating.'	Yi-yor mı-sun? eat-PROG Q-2SG 'Are you eating?'
3	Ye-di-n. eat-PST-2SG 'You ate.'	Ye-di-n mi? eat-PST-2SG Q 'Did you eat?'
4	Ye-di-ler. eat-PST-3PL 'They ate.'	Ye-di-ler mi? eat-PST-3PL Q 'Did they eat?'
5	Kedi yi-yor. cat eat-PROG 'The cat is eating.'	Kedi yi-yor mu? cat eat-PROG Q 'Is the cat eating?'
6	Ali Ayşe ile yi-yor. Ali Ayşe with eat-PROG 'Ali is eating with Ayşe.'	Ali Ayşe ile yi-yor mu? Ali Ayşe with eat-PROG Q 'Is Ali eating with Ayşe?'

Table 8: Statements and Corresponding Yes/No Questions

Based on the data above, the particle is generally added to the end of a declarative sentence to transform it into its interrogative form. However, when asking about the first or second person singular forms in the progressive tense (as demonstrated by examples #1 and #2 in Table 8), the particle instead appears before (and attaches to) the subject agreement suffix. Fur-

ther data would be required to determine if these are the only cases where the particle is not sentence-final.

4.3.2 Wh-Questions

Turkish is wh-in-situ.

#	Statement	Wh-Question
1	Ali pilav yi-yor. Ali rice eat-PROG 'Ali is eating rice.'	Ali ne yi-yor? Ali what eat-PROG 'What is Ali eating?'
2	Ali ev-im-de yi-yor. Ali house-1SG.POSS-LOC eat-PROG 'Ali is eating in my house.'	Ali nerede yi-yor? Ali where eat-PROG 'Where is Ali eating?'
3	Ali Ayşe ile yi-yor. Ali Ayşe with eat-PROG 'Ali is eating with Ayşe.'	Ali kim ile yi-yor? Ali who with eat-PROG 'Who is Ali eating with?'

Table 9: Statements and Corresponding Wh- Questions

The data above shows that wh-words (such as ne 'what') appear in the same position as their corresponding constituents in its declarative form. This suggests that wh-questions are formed purely by substitution of a wh-word and does not involve any movement, supporting the claim that Turkish is a wh-in-situ language.

5 Research Proposal

A major puzzle that remains from my analysis is the phonological process of vowel harmony and consonant assimilation, as discussed in Section 2.5. In short, these processes lead to morphemes appearing in different forms depending on their phonological environment. For example, as mentioned in Section 4.2.1, the locative case suffix can take the form -da/-de/-ta/-te, showing variation that reflects both vowel harmony and consonant assimilation.

This is particularly interesting to me because I am curious whether there are consistent and fully predictable rules governing these processes, which could not be determined from my limited dataset. As for vowel harmony, several specific patterns were observed based on my relatively small dataset, but I was not able to identify a single overarching rule. Consonant assimilation, on the other hand, was too rarely encountered in my dataset, since it appears to only apply to suffixes beginning with plosives or affricates. As such, the limited examples were insufficient to analyze its variation.

If I were to investigate this further, I would first systematically compile a set of suffixes, as well as a wide range of root words that varied in terms of their vowels and consonants. Then, from working with my consultant, I would note the result of attaching each suffix to each root. I would also ask my consultant if multiple suffix forms were ever acceptable.

In the analysis stage, I would isolate vowel harmony effects, focusing on how the vowels of the suffix and root interact. Then, I would separately analyze consonant assimilation, focusing on the environment that determines

voicing alternations.

Overall, this approach would allow for a more systematic investigation of Turkish phonological processes. In particular, a larger and more varied dataset would allow for better identification of underlying patterns.

6 Summary & Conclusion

This paper examined the sound system, morphology and syntax of Turkish based on the fieldwork data I collected. I propose a rich inventory of phones I encountered in the language, and argue that voicing and vowel length are contrastive features. Accordingly, I propose a phonemic inventory consisting of 20 consonant phonemes and 8 vowel phonemes. I then hypothesize phonotactic constraints, specifically a restriction against diphthongs and word-final voiced plosives. Vowel harmony and consonant assimilation are also briefly discussed. In the morphological analysis of this paper, I describe the heavy use of suffixation in Turkish, and provide examples of inflectional and derivational suffixes. Furthermore, an order of suffixes attaching to verbs was identified, and it was noted that allomorphy is common. In the syntactic analysis, Turkish is shown to be a head-final language that follows SOV word order. The formation of interrogative sentences is also discussed.

However, the analysis remains inconclusive in several aspects. In regards to the sound system, I suspect that the inventory of phones and phonemes is incomplete and that the proposed vowel chart is not perfectly accurate, especially due to my limited dataset. In addition to refining the phonemic inventory, future research should also focus on gaining a deeper understanding of vowel harmony and consonant assimilation, as my analysis failed to establish consistent rules. From a morphological perspective, only suffixes were observed, so the possibility of prefixes and infixes should be explored. I also did not investigate the order of multiple derivational morphemes, leaving another area for future research. Syntactically, the head directionality anal-

ysis warrants further investigation, particularly given the challenges posed by Turkish's rich morphology

7 Appendix

Listed below is an organized lexicon of all morphemes I have discovered. Note that allophones/allomorphs are not listed.

7.1 Free Morphemes

Nouns

Transcription	English Gloss
[ʃarku]	‘song’
[kitap]	‘book’
[futbol]	‘football’
[an:ɛ]	‘mother’
[baba]	‘father’
[karɛʃ]	‘sibling’
[abla]	‘older sister’
[abi]	‘older brother’
[tʃɔdʒok]	‘child’
[ɔyul]	‘son’
[kuɪz]	‘daughter’/‘girl’
[gɜz]	‘eye’
[burun]	‘nose’
[ayaz]	‘mouth’
[jʊz]	‘face’
[satʃ]	‘hair’
[ɛl]	‘hand’
[kol]	‘arm’
[ajɛk]	‘foot’
[badʒak]	‘leg’
[karən]	‘stomach’

Transcription	English Gloss
[kɛdiɛ]	‘cat’
[køpɛk]	‘dog’
[at]	‘horse’
[ɛʃɛk]	‘donkey’
[kɛtʃɪ]	‘goat’
[inɛk]	‘cow’
[tavuk]	‘chicken’
[balək]	‘fish’
[faɛ]	‘mouse’
[dʒɛkɛt]	‘jacket’
[masa]	‘table’
[sandaljɛ]	‘chair’
[tɛlɛfɔn]	‘phone’
[ʊst]	‘shirt’
[bilɡɛsajaɾ]	‘computer’
[kalɛm]	‘pen’
[su]	‘water’
[kapɾə]	‘door’
[kaʃik]	‘spoon’
[tʃatal]	‘fork’
[pilav]	‘rice’
[ɛv]	‘house’
[jatak]	‘bed’
[kəs]	‘decrease’
[tʊmɛn]	‘group of’
[dʊmɛn]	‘ship steering wheel’
[kaz]	‘goose’
[gaz]	‘gas’
[kan]	‘blood’
[hala]	‘aunt’
[haːla]	‘still’

Transcription	English Gloss
[kaɾ]	‘snow’
[ka:ɾ]	‘profit’
[a:dɛt]	‘custom/tradition’

Verbs

Transcription	English Gloss
[oku]	‘read’
[konuʃ]	‘speak’
[søjlɛ]	‘say’
[jɛ]	‘eat’
[jap]	‘do’
[git]	‘go’
[øɫ]	‘die’
[uju]	‘sleep’
[iste]	‘want’
[ø:rɛn]	‘learn’
[ø:rɛt]	‘teach’
[ɐɭɐ]	‘cry’
[tʃʊlʊkat]	‘scream’
[sɛv]	‘love’
[nɛfrɛtɛt]	‘hate’
[bile]	‘know’
[dʊsʊn]	‘think’

Adjectives

Transcription	English Gloss
[kʊtʃʊk]	‘small’
[bʊjʊk]	‘big’

Transcription	English Gloss
[atʃ]	‘hungry’
[ije]	‘good’
[køtʊ]	‘bad’
[motlu]	‘happy’
[ʌzgʊn]	‘sad’
[jorgʊn]	‘tired’
[kəmɪk]	‘funny’
[havalə]	‘cool’
[zɛke]	‘smart’
[sɔyuk]	‘cold’
[uzun]	‘tall’
[kəsa]	‘short’
[a:ma:]	‘blind’

Others

Transcription	English Gloss	Category
[biɣ]	‘one’	numeral
[iki]	‘two’	numeral
[ʌtʃ]	‘three’	numeral
[dørt]	‘four’	numeral
[bɛf]	‘five’	numeral
[altə]	‘six’	numeral
[jɛde]	‘seven’	numeral
[sekɪz]	‘eight’	numeral
[dɔkuz]	‘nine’	numeral
[ɔn]	‘ten’	numeral
[bɛn]	‘I’	pronoun
[sɛn]	‘you (sg)’	pronoun
[o]	‘he/she/it’	pronoun

Transcription	English Gloss	Category
[biz]	‘we’	pronoun
[siz]	‘you (pl/formal)’	pronoun
[ama]	‘but’	conjunction
[ilɛ]	‘with’	postposition
[hakənda]	‘about’	postposition
[mi]	‘yes/no’	question marker
[nɛ]	‘what’	question word
[nɛɾɛdɛ]	‘where’	question word
[kɪm]	‘who’	question word

7.2 Bound Morphemes

Transcription	Attaches to	Function / Meaning
[-di]	verb	past tense marker
[-jɔɾ]	verb	progressive aspect marker
[-edʒɛk]	verb	future tense marker
[-r]	verb	present/habitual marker
[-m]	verb / noun	1SG subject agreement
[-n]	verb / noun	2SG subject agreement
[-siniz]	verb	2PL subject agreement
[-lɛr]	noun / verb	plural / 3PL marker
[-mɛ]	verb	verbal negation marker
[-dʒi]	noun	agentive suffix
[-dɪr]	verb	causative marker
[-mɛk]	verb	infinitive marker
[-in]	noun	genitive case marker
[-siz]	noun	without noun
[-le]	noun	with noun
[-mɛk]	verb	infinitive marker
[-dʊʏ]	verb	complementizer

Transcription	Attaches to	Function / Meaning
[-i]	noun	accusative case marker
[-dɛ]	noun	locative case

References

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